



OBJECT ORIENTED ANALYSIS AND DESIGN

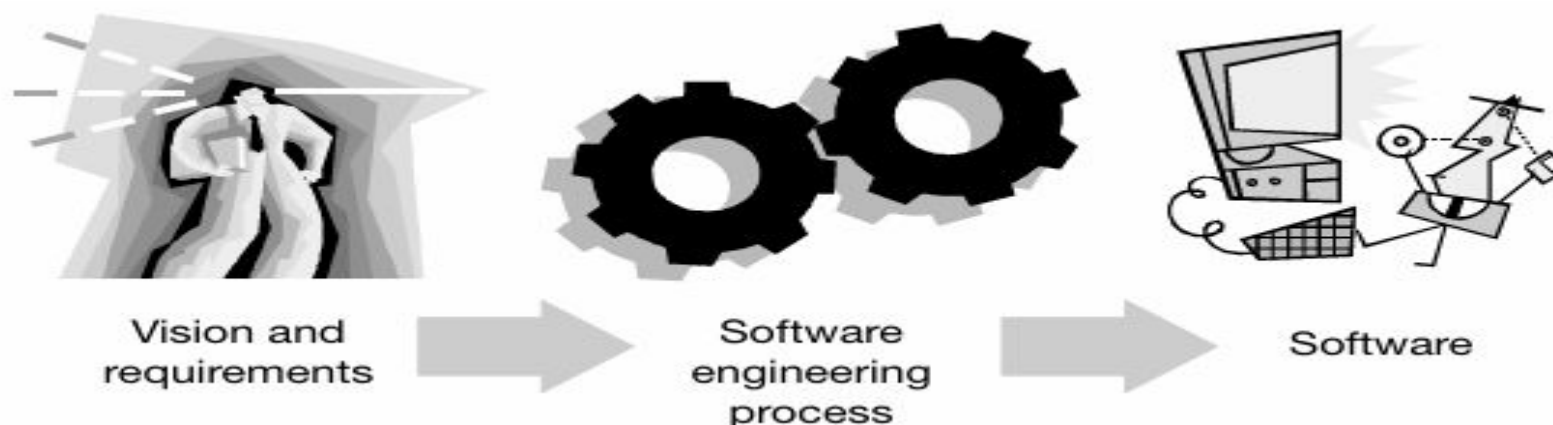
Ch - 2

A study material for the students of GLS University

What is Unified Process?

- The Unified Process (UP) is a method or way of developing software in a well-planned and organized manner.
- It helps software teams to:
- Understand what to build,
- Design it properly,
- Build it step by step,
- And deliver it to users successfully.

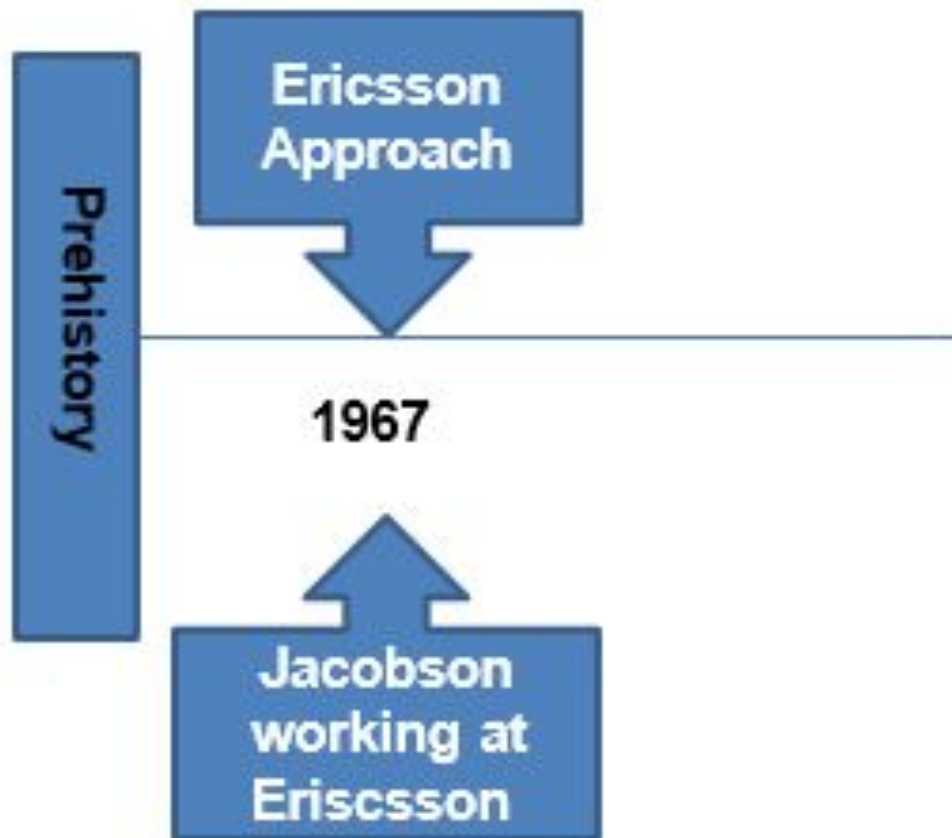
SEP is the process in which user requirements convert into software



- UML project was meant to provide both visual language and software engineering process
- UML is visual language and UP is the process part
- UP is based on process work conducted by Ericsson approach at Rational
- UP method for developing software and incorporates best practice from its predecessors

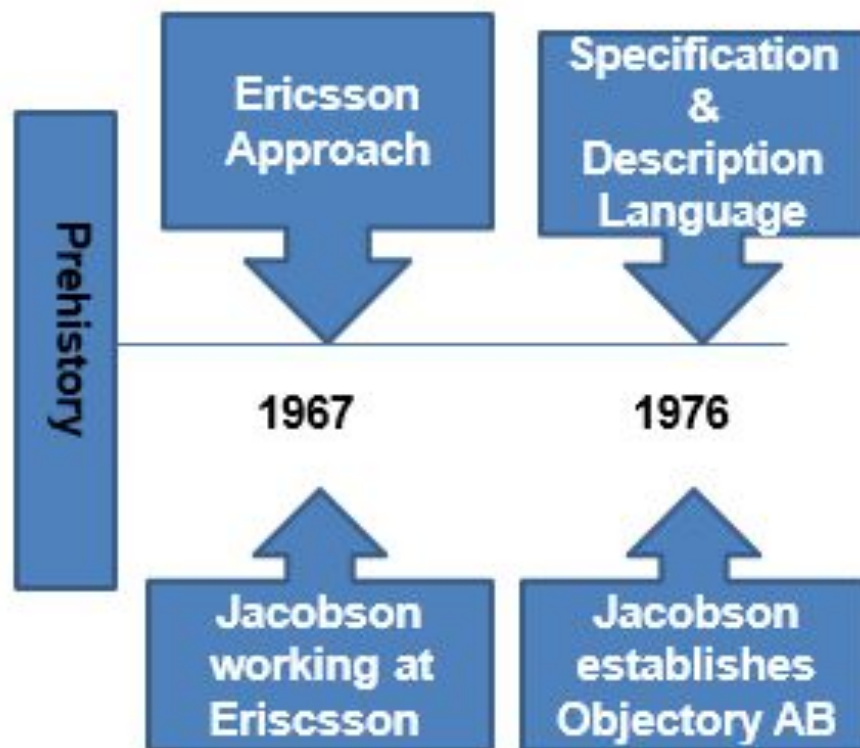
The Birth of UP

- Ivar Jacobson, father of UP
- In 1967 the **Ericsson Approach**, modeling a complex system as a set of interconnected blocks
- Small blocks were interconnected to form large blocks building up to a complete system
- The basis of this approach was “Divide and conquer” and now as component-based development

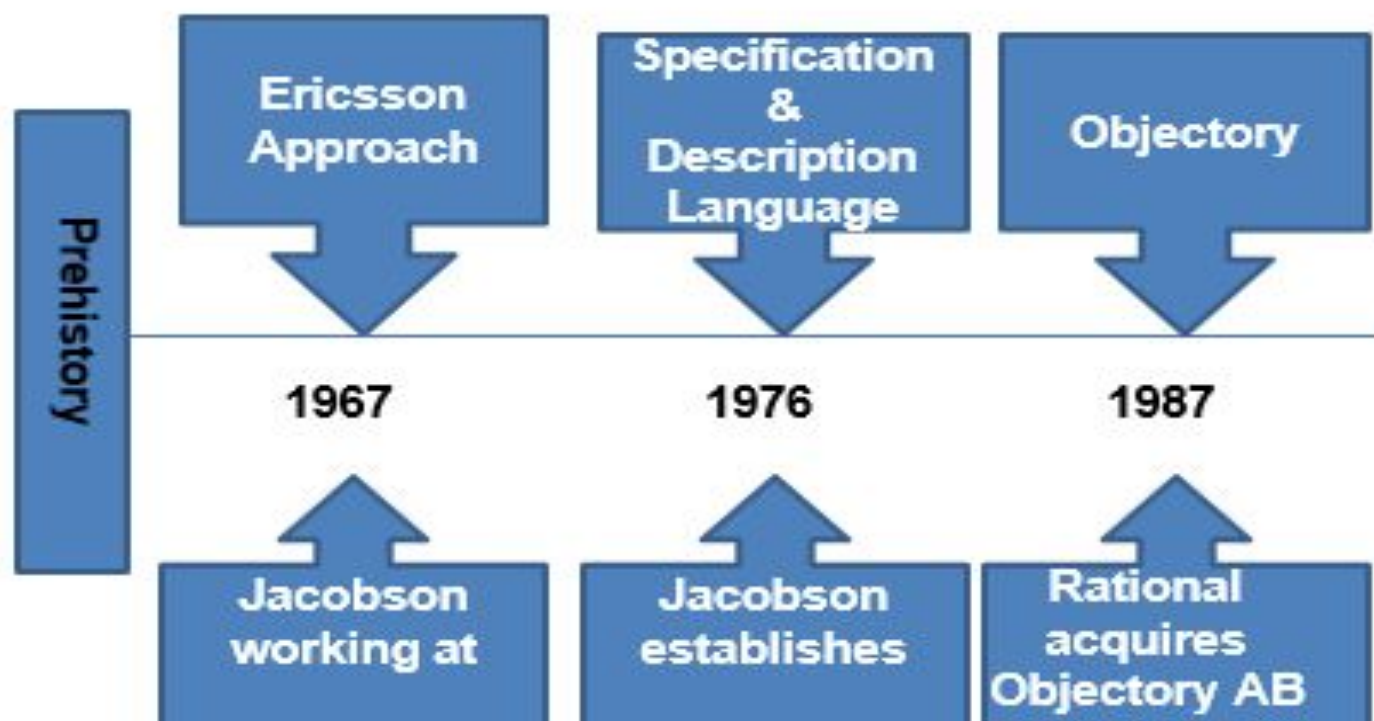


- Another Ericsson innovation was, these blocks creating “traffic cases” that described how system to be used – which called use cases in UML
- The result of this process was an architecture representation that described the blocks and how they fitted together
- The requirement view (traffic cases) and static view (architecture description), Ericsson had a dynamic view that described how all the blocks communicated with each other
- This consist of sequence, communication and state machine diagram

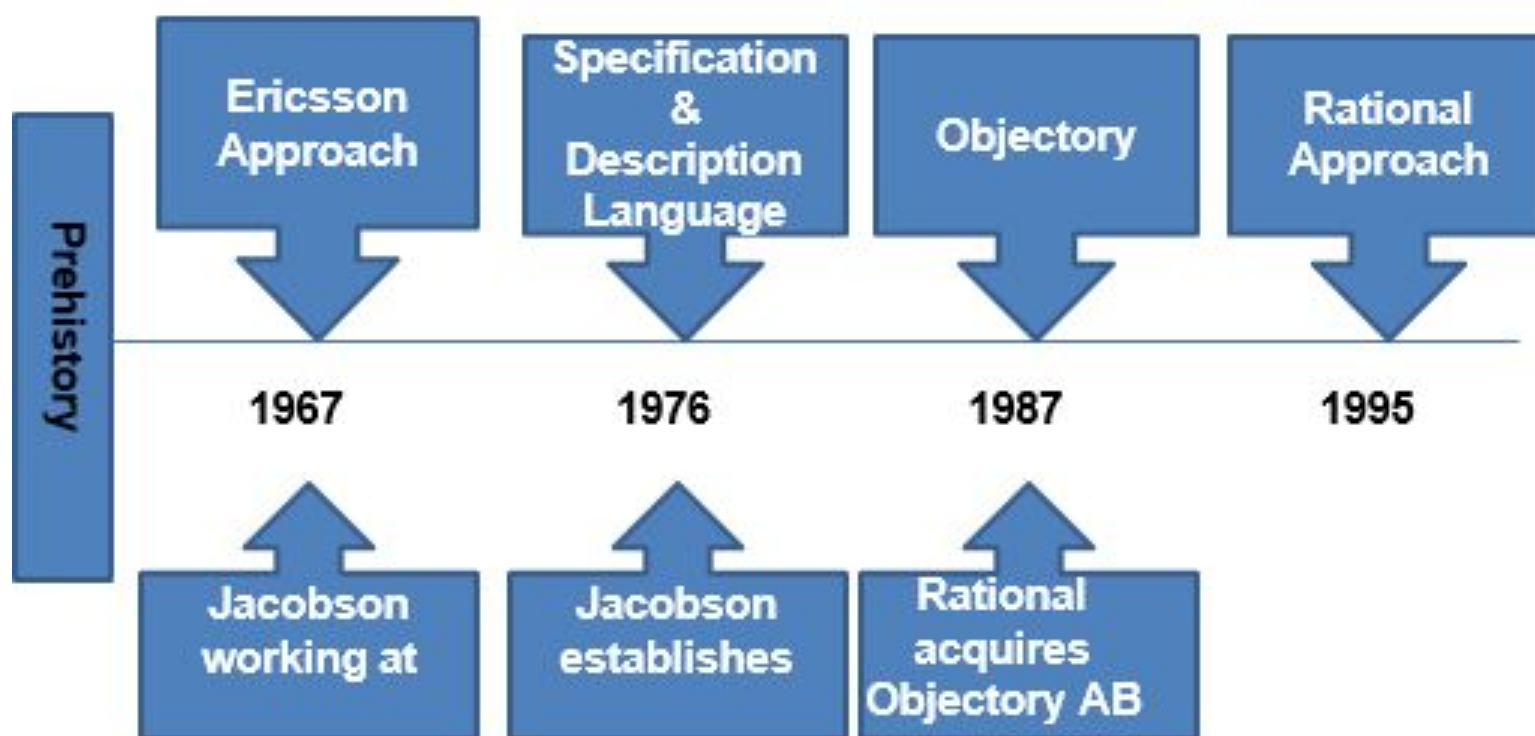
- Next development in 1980s of Specification & Description Language (SDL) was one of the first object-based visual modeling language
- In 1992 it was extended to become object-oriented with classes and inheritance



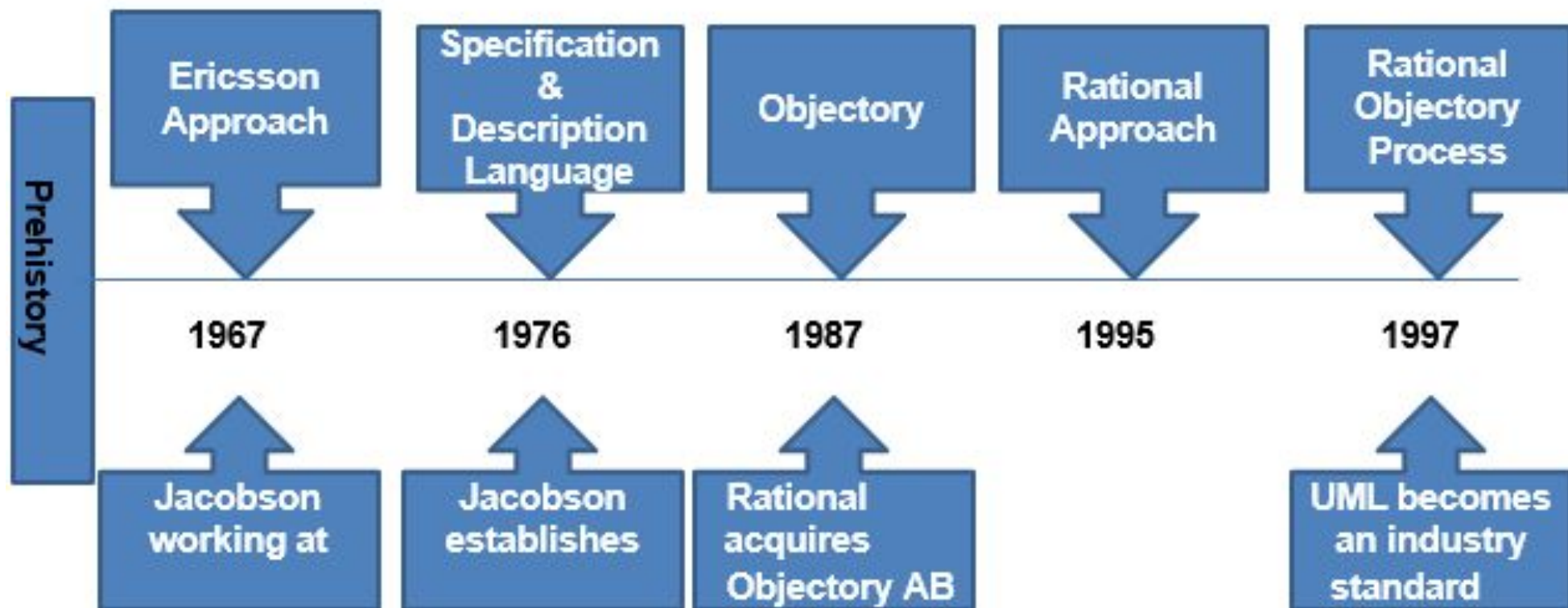
- In 1987, company develop and sold a software engineering process, based on Ericsson approach called Objectory
- It basically consist of Documentation and tools needed by objectory



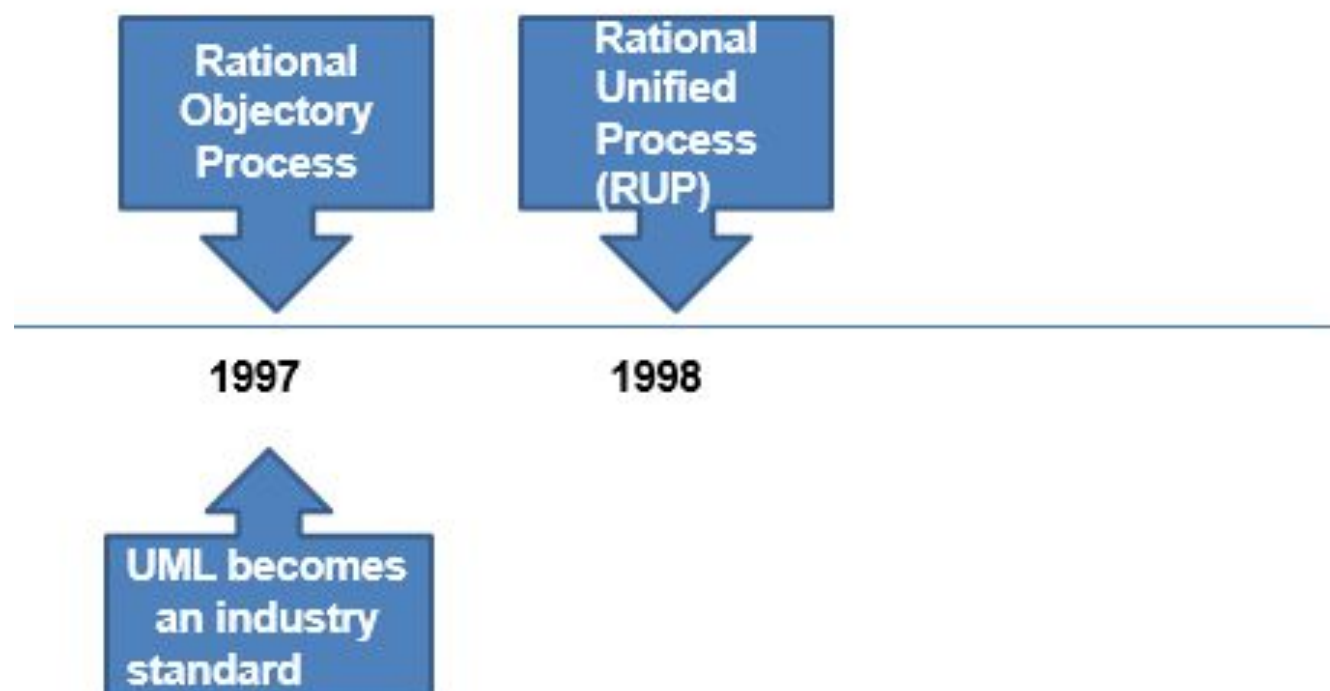
- Objectory acquired by Rational in 1995, objectory process with large amount of process related work that already been done by rational
- “4+1 view” was developed



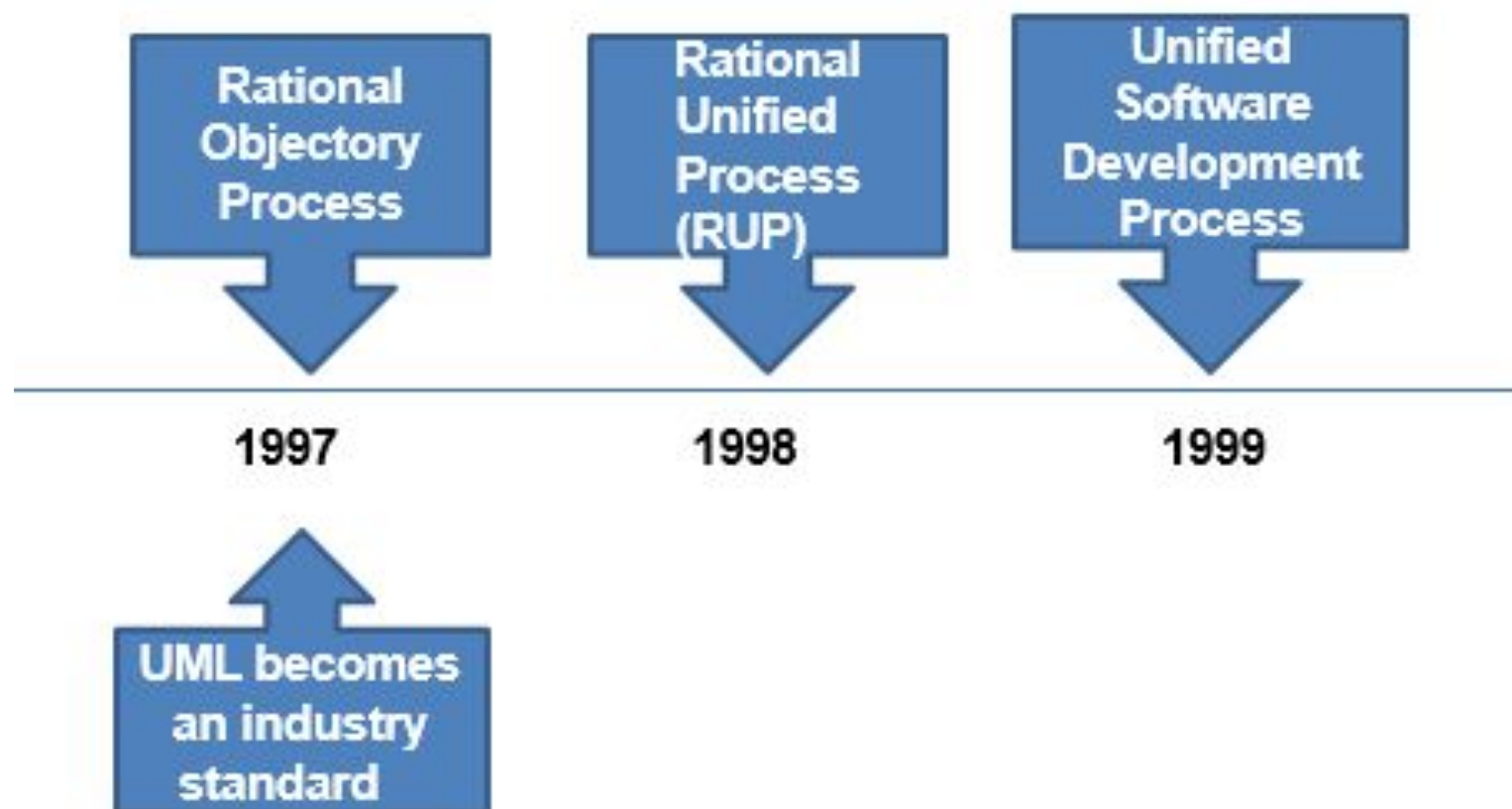
- The Rational Objectory Process (ROP) was the result of the unification of Objectory with Rational's process work



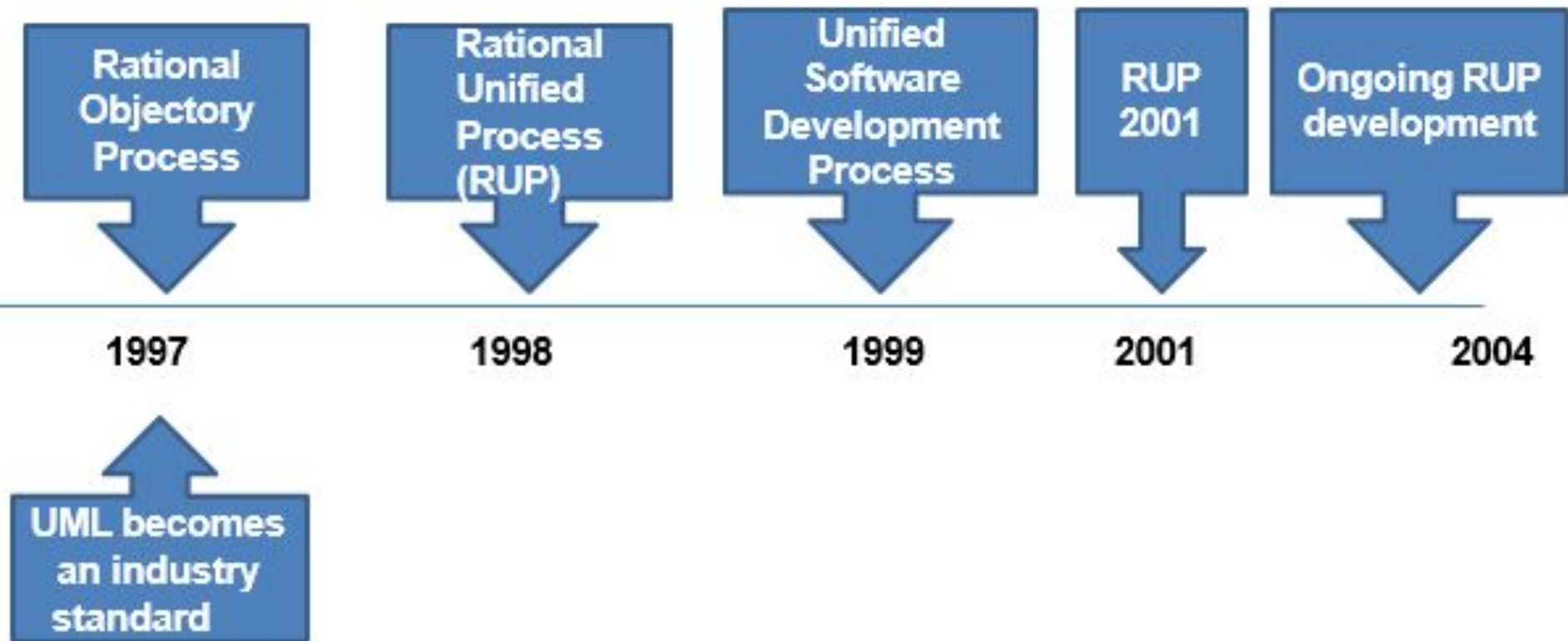
- In 1997, Rational acquired many more companies bringing in expertise in requirements capture, configuration management, testing
- This led to the release of Rational Unified Process (RUP) in 1998



- In 1999, the Unified Software Development Process, which describes the Unified Process



- RUP is Rational Process Product and UP is an open SEP Software Engineering Process
- Both are closely related



UP Axioms

UP axioms are the fundamental principles or guiding rules of the Unified Process (UP).

1. Use-Case Driven
2. Architecture-Centric
3. Iterative and Incremental
4. Risk Focused
5. Model Driven

1. Use-Case Driven –

Use cases help define system behavior and ensure that development focuses on fulfilling user needs.

All development activities—from design to testing—are tied back to use cases.

2. Architecture-Centric –

How system divided into components

How those components interact

How Deployed on hardware

3. Iterative and Incremental –

Iterative, break the project into small subprojects that deliver system functionality

- Increments, leading to a fully functional system or we build software by a process of stepwise refinement to our goal

4. Risk-Focused –

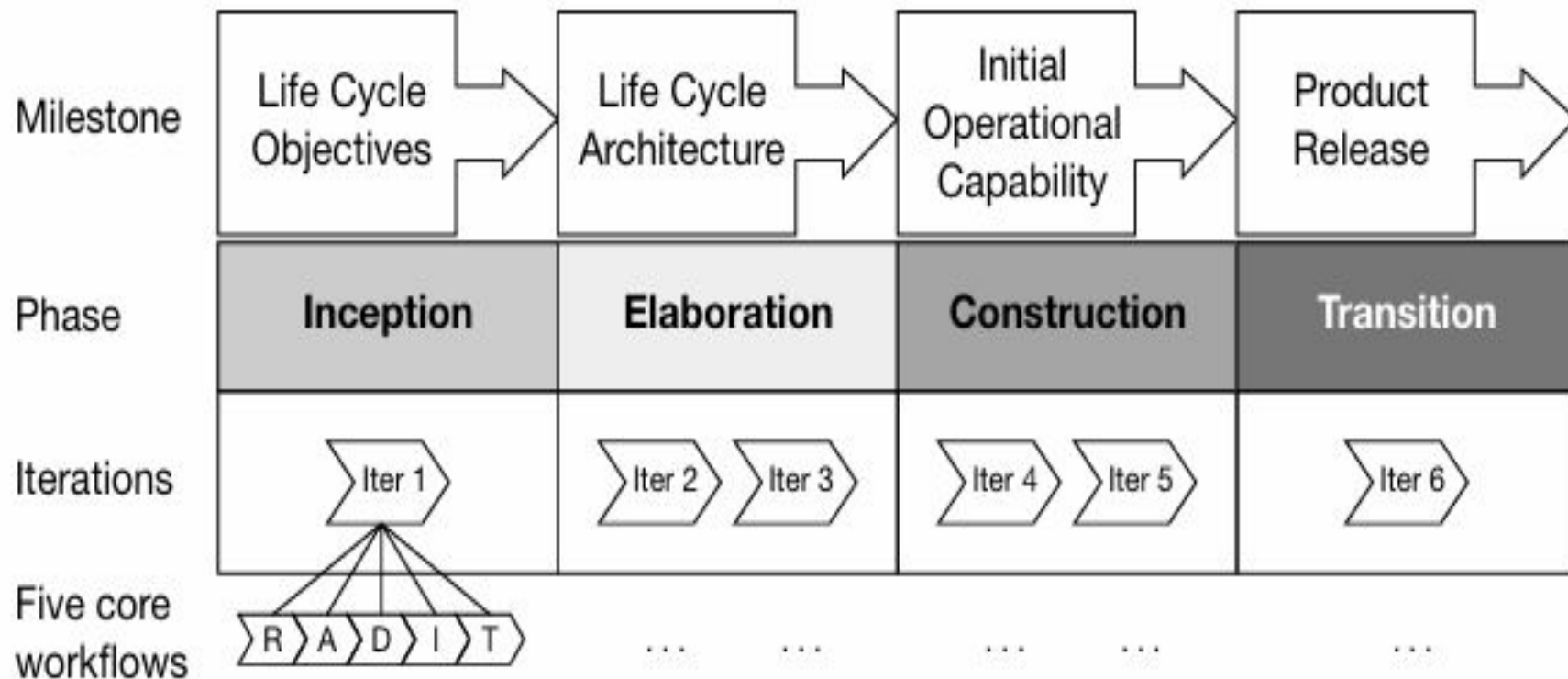
Early identification and mitigation of risks are key to success.

5. Model Driven –

Models guide coding, documentation, and communication among team members.

Emphasizes the use of visual models (e.g., UML diagrams) to represent system structure and behavior.

UP Structure



UP Structure

The product life cycle is divided into four phases

Inception, Elaboration, Construction and Transition -
Each of which ends with major milestone

- Each phase contains one or more iterations
- Each iteration execute the five core workflows and any extra workflows

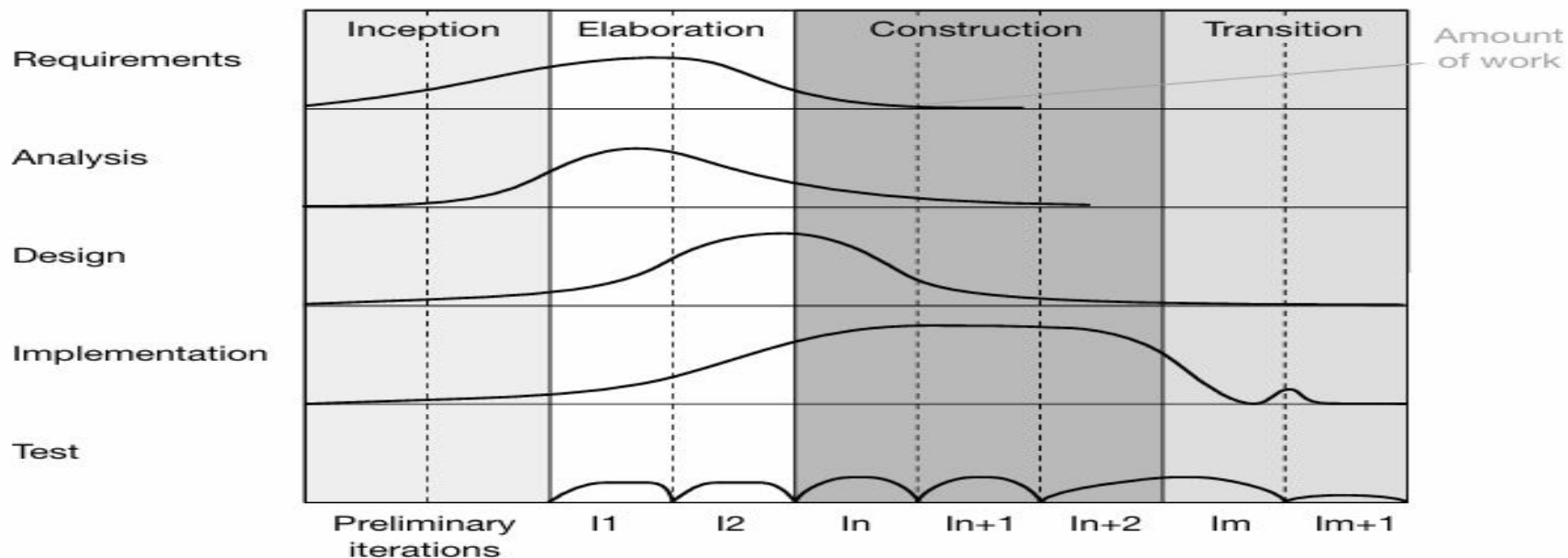
UP Structure

UP consists of a sequence of four phases, each of which terminates with a major milestone:

- 1. Inception – Life Cycle Objectives**
- 2. Elaboration – Life Cycle Architecture**
- 3. Construction – Initial Operational Capability**
- 4. Transition – Product release**

Core Workflows (R-A-D-I-T) :

Workflow	Focus
R – Requirements	Define what the system should do
A – Analysis	Understand the domain and refine requirements
D – Design	Define the architecture and system design
I – Implementation	Coding and integrating components
T – Test	Ensure system correctness and quality



Workload Variation by Phase

Phase	Main Workflows
Inception	Heavy focus on Requirements & Analysis
Elaboration	Strong in Requirements, Analysis, some Design
Construction	Strong in Design & Implementation
Transition	Focus on Implementation & Testing

UP Phases

1. **Inception Phase** – Inception is all about initiating the project
 - Initiation involves establishing feasibility, creating a business case, capturing requirement and identifying critical risks
 - Primary workers - project manager and system architect
 - Focus on requirement and analysis workflow

2. Elaboration Phase –

Elaboration is about creating a partial but working version of the system – an **executable architecture baseline**

- Main goals
 - Create an executable architecture baseline
 - Create a detailed plan for the construction phase
 - Formulate resources, time, equipment, staff and cost
- Main Focus on Design workflow and other also refer

3. Construction Phase –

Construction evolves the executable architectural baseline into a complete, working system

- Goal of construction is to complete all requirement, analysis, and design and to evolve the architectural baseline generated in Elaboration into the final system
- Main focus on the implementation workflow
- System is ready for testing at user site

4. Transition Phase –

Transition is about deploying the completed system into the user community

- This phase starts when testing is completed and the system is finally deployed
- Goals of this phase – prepare the user sites for the new software, modify the software, create user manuals and other documentation
- Main focus on test workflow

iterative and incremental process

The Unified Process (UP) is fundamentally an iterative and incremental software development framework designed to tackle complex projects by breaking them into manageable parts and evolving the system step by step.

Iterative -

development in repeatable cycles (iterations).

Each cycle encompasses **planning, analysis, design, implementation, integration, test, and often a release.**

After each cycle, you review, refine, and improve the work.

Ex- drafting multiple versions of an essay—refine at each version.

Baseline -

- Each iteration generates a baseline
- That Comprises a partially complete version of the final system and any associated project documentation
- Baselines build on each other over successive iterations until the final finished system is achieved

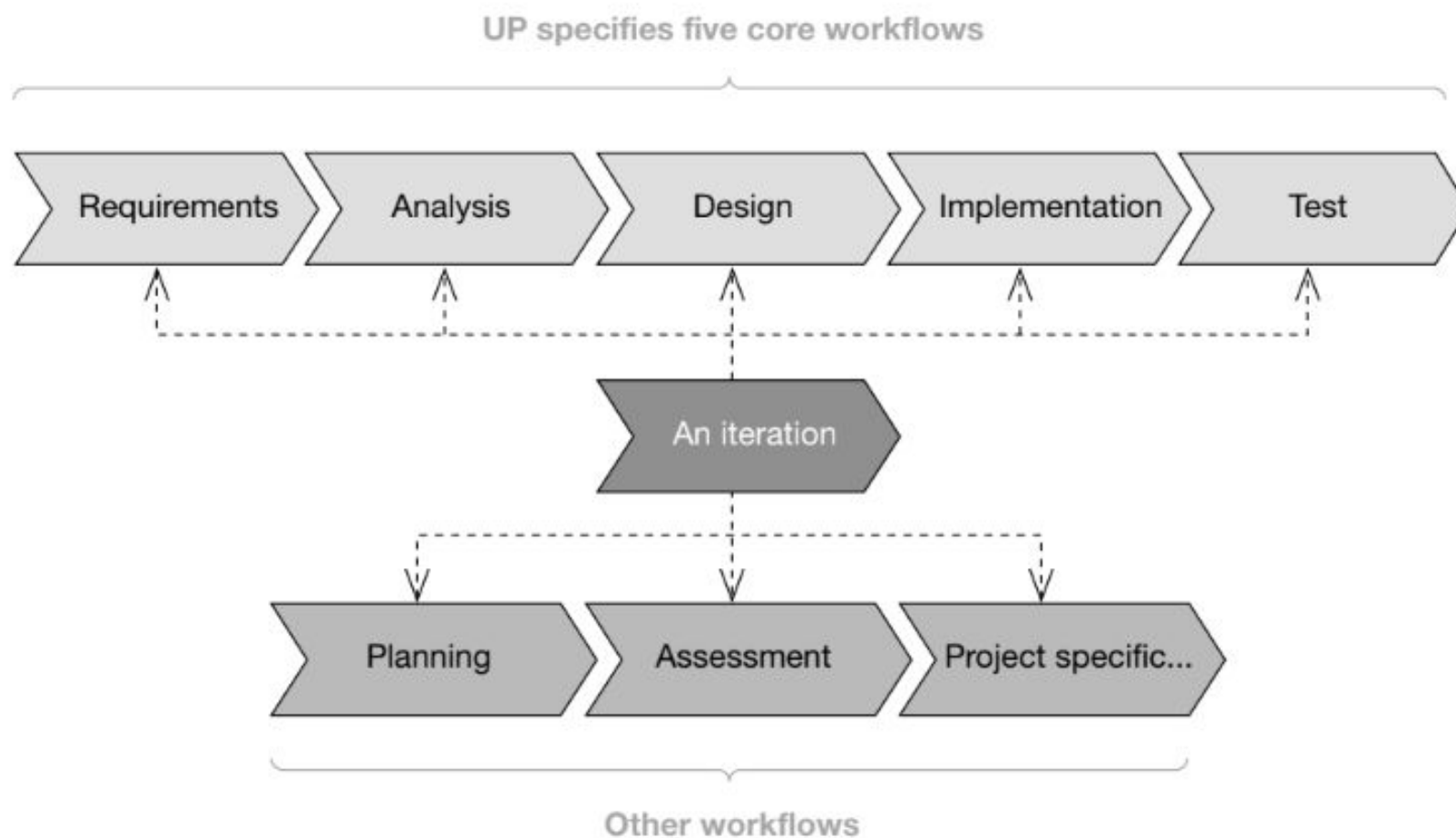
Incremental -
building the system piece by piece.

Each new “increment” adds new functionality to the previous baseline.

Over time, these increments sum up to a complete system.

The difference between two consecutive baselines is known as an increment

Iteration Workflows -



Iterations & Baselines -

- Each iteration produces a baseline—a partially complete version of the system, along with documentation.
- Each baseline provides an agreed basis for further review and development
- Can be changed only through formal procedures of configuration and change management
- Increments are just the difference between one baseline and the next

Think of baseline as a saved game. If something goes wrong, you can reload it. And incremental means you keep upgrading your game level by level, without restarting.

MCQs

1. Which phase of UP defines the system architecture?

- a) Inception
- b) Elaboration
- c) Construction
- d) Transition

Answer: b) Elaboration

2. The main goal of the Inception phase is:

- a) To deliver the final product
- b) To capture 100% of requirements
- c) To check feasibility and define scope
- d) To write complete code

Answer: c) To check feasibility and define scope

MCQs

3. In which UP phase is the software deployed to the user environment?

- a) Construction
- b) Transition
- c) Inception
- d) Elaboration

Answer: b) Transition

4. Which of the following is a milestone of the Construction phase?

- a) Product Release
- b) Life Cycle Objectives
- c) Initial Operational Capability
- d) Risk Resolution

Answer: c) Initial Operational Capability

MCQs

5. What is the main workflow focus in the Construction phase?

- a) Requirement Gathering
- b) System Design
- c) Implementation & Testing
- d) Deployment

Answer: c) Implementation & Testing

6. The final milestone of the Unified Process is:

- a) Life Cycle Architecture
- b) Product Release
- c) Initial Operational Capability
- d) Vision Statement

Answer: b) Product Release

The Unified Process has three phases: Inception, Construction, and Deployment.

False

Elaboration phase ends with the milestone called "Life Cycle Architecture".

True

Requirements are completely frozen after the Inception phase.

False

The Transition phase involves testing and bug fixing.

True

1. Who is known as the father of the Unified Process?

Answer : Ivar Jacobson

2. Which phase includes most of the coding and testing?

Answer: Construction

3. Milestone achieved at the end of Elaboration?

Answer: Life Cycle Architecture

4. Final milestone of the UP?

Answer: Product Release

Real-World Analogy Exercise

Question:

Imagine you are someone responsible for the organizing of a college fest at your own campus or in the planning of a birthday party for one friend. Software development, likewise to the two events, needs planning, also coordination, plus execution.

Use your Unified Process understanding to explain how you would manage each of its four phases when you organize the event: Inception, Elaboration, Construction, Transition.

In your answer, describe:

- In each phase, what activities or decisions take place then?
- What type of information would you want to collect?
- What are some problems you might anticipate and also how you make plans in order to deal with them
- Each phase prepares you. That groundwork is what the following stage develops.

Opening a New Restaurant – Understanding UP Phases

Phase 1: Inception = Initial Idea & Business Plan

What type of restaurant? (fast food, fine dining, cafe?)

Who are your customers?

What's your budget?

What's your vision?

Phase 2: Elaboration = Designing & Planning

Finalize the menu

Design the kitchen and dining area

Plan staff and equipment

Handle safety and legal risks

Opening a New Restaurant – Understanding UP Phases

Phase 3: Construction = Building the Restaurant

Hire chefs and staff

Purchase equipment

Decorate the space

Test kitchen equipment and sample dishes

Phase 4: Transition = Opening & Feedback

Grand opening

Gather customer feedback

Fix service issues

Launch advertising

THANK YOU